

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Electronic Device and Circuits (New)

Pass Marks: 45

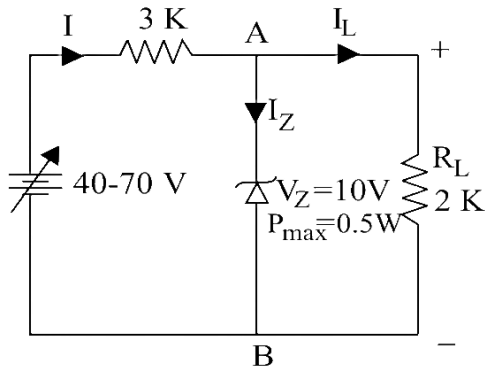
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

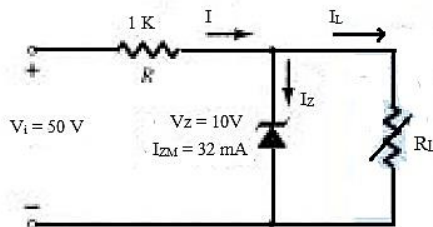
Attempt all the questions.

1. a) Briefly discuss the doping of semiconductor materials and the construction of semiconductor diode. 7
- b) Calculate the battery current I , I_Z and I_L in the circuit shown. How will these values be affected if source voltage increases to 70 V? Neglect Zener resistance. 8

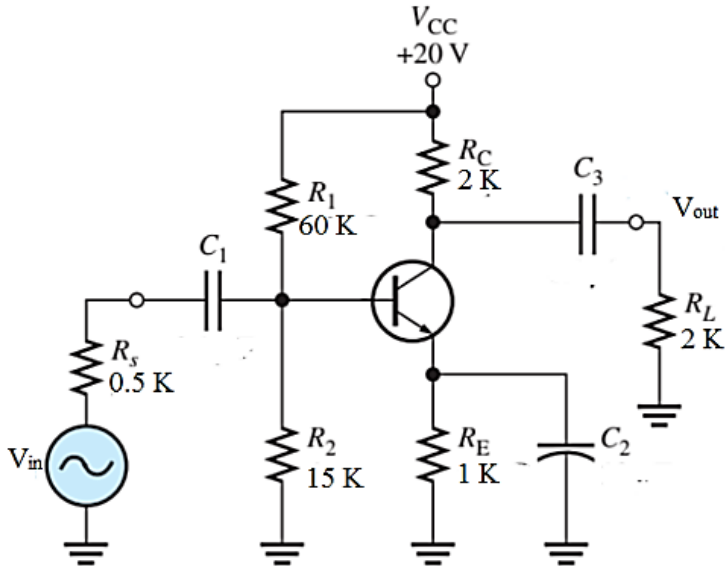


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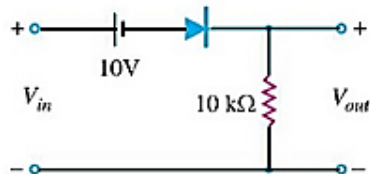
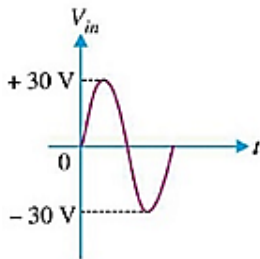
For the network of figure below, determine the range of R_L and I_L that will maintain the output voltage at 10 V. Also determine the maximum wattage rating of the diode.



2. a) Explain the common emitter (CE) configuration of BJT with necessary circuit diagram. Draw the input and output characteristics and indicate different regions of operations. 7
- b) For the circuit shown in figure below draw dc load line and locate operating point. Assume $\beta=50$. 8

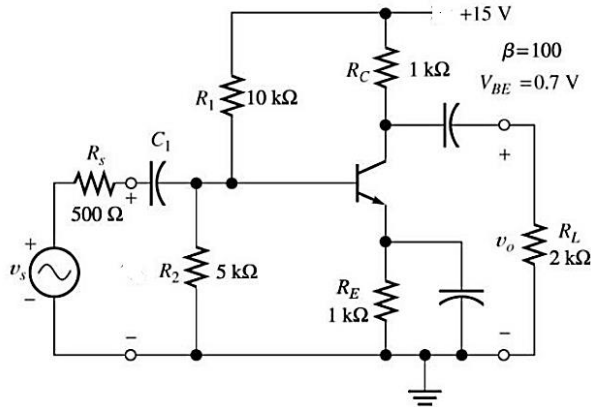


3. a) Explain the working principle of full wave bridge rectifier. Also, derive the expression for efficiency of full wave bridge rectifier. 8
- b) What are the clipper and clamper circuits? For the circuit shown in figure below, determine the output waveform and also explain how it is obtained. 7



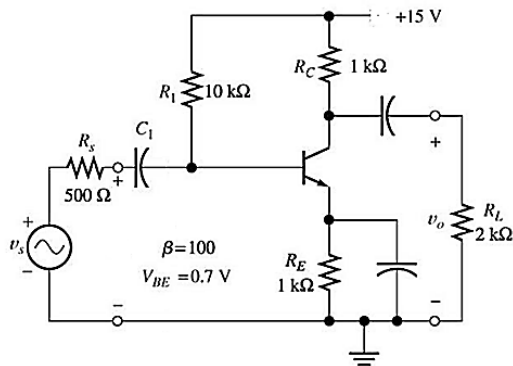
4. a) Explain the construction and working principle of D-MOSFET. 7

- b) Draw r_e model and find input impedance, output impedance and overall voltage gain. 8



OR

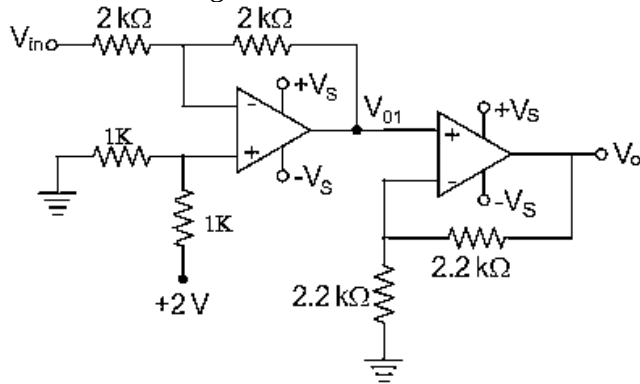
Draw r_e model and find input impedance, output impedance and overall voltage gain.



5. a) Derive an expression of overall voltage gain for multistage amplifier. 8
The voltage gain of first, second, third and fourth stages in multistage amplifier are -24, -60, -12 and 0.9 respectively. Find the voltage gain (in dB) of fifth stage, which, if added to the cascade, would make the overall voltage gain 10^6 .
- b) What are the different classes of amplifiers? Draw transformer coupled push pull amplifier and derive an expression of its efficiency. 7
6. a) Explain the effect of negative feedback in bandwidth? An amplifier has the midband gain of 1500 and a bandwidth of 4 MHz. The midband gain reduces to 150 when negative feedback is applied. Determine the value of feedback factor and bandwidth. 7

b) For the circuit shown in figure below, if $V_{in} = -3.5\text{ V}$ find V_o .

8



7. Write short notes on: (**Any two**)

2×5

- a) Multi vibrator
- b) BJT switching times
- c) Common collector configuration